

Problem 5.14

On January 1, 2000, you invested \$1,000. Your investment grows to \$1,400 by December 31, 2007. What was the exact simple interest rate at which you invested?

The total interest earned is

$$I = 1400 - 1000 = 400.$$

We use the simple interest formula:

$$I = Prt,$$

where $P = 1000$, $I = 400$, and t is the time in years.

Time

From January 1, 2000 to December 31, 2007, the total number of days is

$$366 + 365 + 365 + 365 + 366 + 365 + 365 + 365 = 2922 \text{ days.}$$

Thus, using exact simple interest,

$$t = \frac{2922}{365}.$$

Solve for the rate

Substituting into $I = Prt$:

$$400 = 1000 \cdot r \cdot \frac{2922}{365}.$$

Solving for r ,

$$r = \frac{400}{1000} \cdot \frac{365}{2922}.$$

Therefore,

$$r \approx 0.04997.$$

So, the exact simple interest rate is

$$\boxed{4.997\% \approx 5\%}.$$